

What Can You Measure from a Single Photo?

Goal: From one RGB image, recover per-pixel

- ▶ **Metric depth** — distance in meters from camera to each point
- ▶ **Surface normals** — 3D orientation of each surface patch

Demo output:
RGB | Depth | Normals
(replace with inference result)

Connection to course: The pinhole camera model (Ch. 2) relates 3D world coordinates to 2D image coordinates via focal length f :

$$u = f \frac{X}{Z}, \quad v = f \frac{Y}{Z}$$

Without knowing f , depth is ambiguous.